

Holistic Data Preparer

Final Project - Part A: Conceptual Foundation

1. What is Data Analysis?

Data Analysis is the process of collecting, cleaning, transforming and analyzing data to discover useful information and support decision-making.

It helps us understand patterns, trends and relationships present in the data.

The main steps are: Data Collection, Data Cleaning, Data Transformation, Data Analysis, Data Visualization, Interpretation and Decision Making.

In this project, Data Analysis is used to understand customer financial information and prepare it for Machine Learning.

2. Data Science Project Planning

A Data Science project follows a structured process:

1. Problem Definition
2. Data Collection
3. Data Understanding
4. Data Cleaning
5. Exploratory Data Analysis
6. Feature Engineering
7. Data Preprocessing
8. Model Building
9. Model Evaluation
10. Deployment and Monitoring

3. Machine Learning Problem Statement

Customer Loan Default Prediction



The objective is to predict whether a customer is likely to default on a loan based on customer and financial information.

Important features include Age, Annual Income, Loan Amount, Credit Score, Repayment History, Transaction Count, Spending Ratio, Employment Type, Education Level, Region and Loan Purpose.

The target variable is `default_flag`:

- 0 = Customer did not default

- 1 = Customer defaulted

This is a Supervised Machine Learning and Binary Classification problem.

4. Project Objective

The main objective is to prepare the Customer Credit Risk dataset for Machine Learning.

The project includes understanding the dataset, identifying missing values, handling outliers, performing EDA, encoding categorical features, scaling numerical features, applying transformations, creating new features, building a preprocessing pipeline and generating a final clean dataset.

5. What are Tensors?

A Tensor is a way of representing numerical data in different dimensions.

The main types are:

- 0D Tensor: Scalar

- 1D Tensor: Vector

- 2D Tensor: Matrix

- 3D Tensor: Collection of Matrices

NumPy can be used to create and understand tensors.

Note: Tensor theory is included here; the Tensor code examples have been removed as requested.

Part A Conclusion

In this section, the conceptual foundation of the project was established.

Topics covered: Data Analysis, Data Science Project Planning, Machine Learning Problem Statement, Customer Loan Default Prediction, Binary Classification, and Tensors with different dimensions.

The next section will focus on Data Acquisition and importing the dataset for further analysis.